

Computer Network

Final Project Report

CSCE 3530.401

University of North Texas

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1. Project Overview

This project implements a small virtual office network using Cisco Packet Tracer, demonstrating how core networking protocols and services work together in a client-server environment. The network includes a router, a Layer 2 switch, two client workstations, a combined DHCP/DNS server, a web server, and a simulated external host representing the public internet.

The goal was to design, configure, test, and document a functional network in which devices communicate correctly using DHCP, DNS, ARP, HTTP/HTTPS, ICMP, TCP, UDP, NAT, and SSL/TLS — the protocol set specified in the project proposal.

2. Objectives of the Project

- Design a small office network topology that realistically reflects a client-server environment, including internal clients, shared services, and a connection point to an external network.
- Configure core network infrastructure (router and switch) using real Cisco IOS command-line syntax, including interface activation, IP addressing, and NAT.
- Deploy and configure DHCP and DNS services so that clients receive addressing information automatically and can resolve hostnames to IP addresses.
- Deploy a web server supporting both HTTP and HTTPS, and verify secure and non-secure communication from a client.
- Demonstrate and verify each required protocol (DHCP, DNS, ARP, HTTP/HTTPS, ICMP, TCP, UDP, NAT, SSL/TLS) through hands-on configuration and testing, capturing evidence of each working correctly.
- Document the design, configuration, and test results in a format suitable for a live demonstration and written report.

3. Network Design

The network follows a star topology centered on a Layer 2 switch, with a router providing the gateway to a simulated external network. All internal devices sit on the 192.168.1.0/24 subnet; the router's outside interface and the simulated internet host sit on a separate 203.0.113.0/24 subnet, consistent with NAT best practice of never overlapping inside/outside address space.

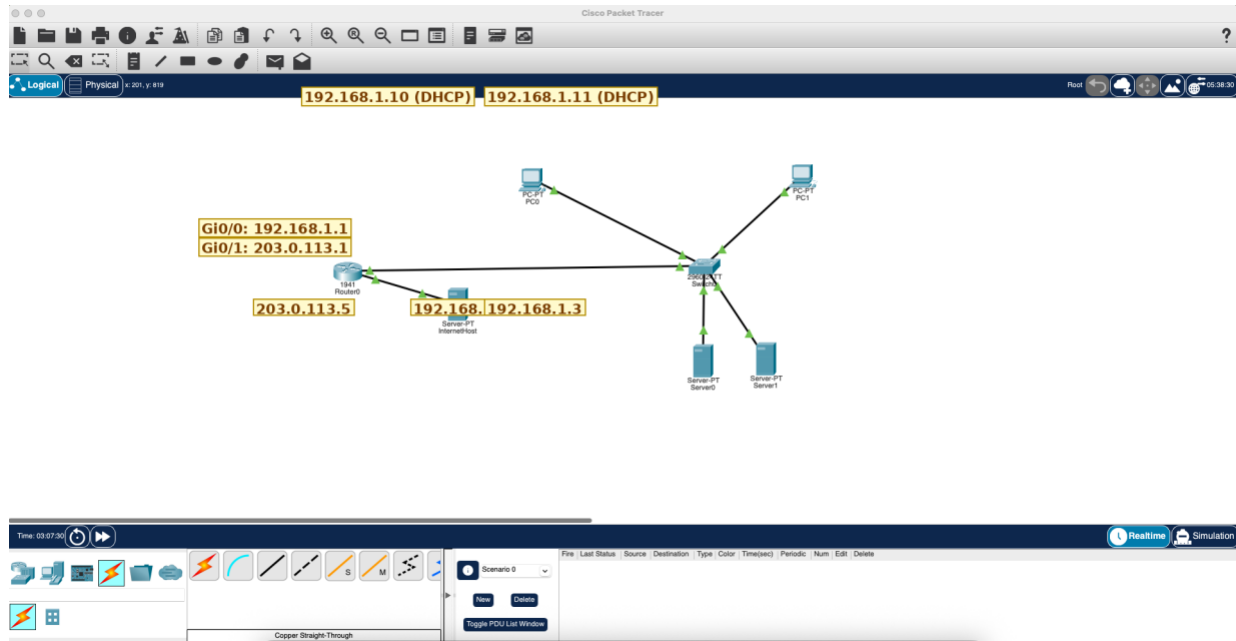


Figure 1. Final network topology with IP addressing labeled directly on each device/link.

3.1 IP Addressing Table

Device	Role	IP Address	Interface
Router0 (Cisco 1941)	Default gateway, NAT translation	192.168.1.1 (LAN) 203.0.113.1 (WAN)	Gi0/0, Gi0/1
Switch0 (Cisco 2960-24TT)	Layer 2 LAN backbone	N/A	Fa0/1-0/4
Server0	DHCP + DNS server	192.168.1.2	FastEthernet0
Server1	Web server (HTTP/HTTPS)	192.168.1.3	FastEthernet0
PC0	DHCP client	192.168.1.10 (leased)	FastEthernet0
PC1	DHCP client	192.168.1.11 (leased)	FastEthernet0
InternetHost	Simulated external / public network	203.0.113.5	FastEthernet0

4. Device Configuration

4.1 Router Configuration

Router0's LAN-facing interface (GigabitEthernet0/0) was activated and assigned the gateway address 192.168.1.1/24. By default, Cisco router interfaces are administratively shut down, so a 'no shutdown' command was required before the interface would pass traffic.

```
Router(config-if)#no shutdown

Router(config-if)#
%LINK-S-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-S-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#
```

Figure 2. Router0 CLI — interface activation and IP addressing.

4.2 Switch Configuration

Switch0 required no IP addressing, since Layer 2 switches forward frames based on MAC address rather than IP address. Its access ports came up automatically once cables were connected, after passing through Spanning Tree Protocol's listening/learning states.

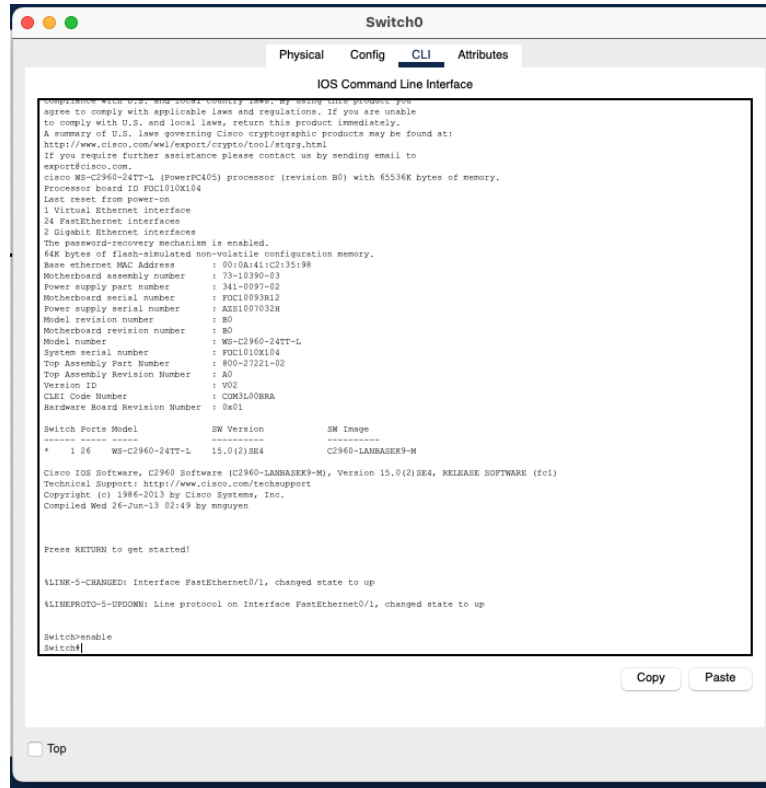


Figure 3. Switch0 CLI confirming interface status.

4.3 Client Configuration

PC0 was initially assigned a static IP address to validate basic Layer 3 connectivity to the router before DHCP was configured, confirming the addressing plan was correct.

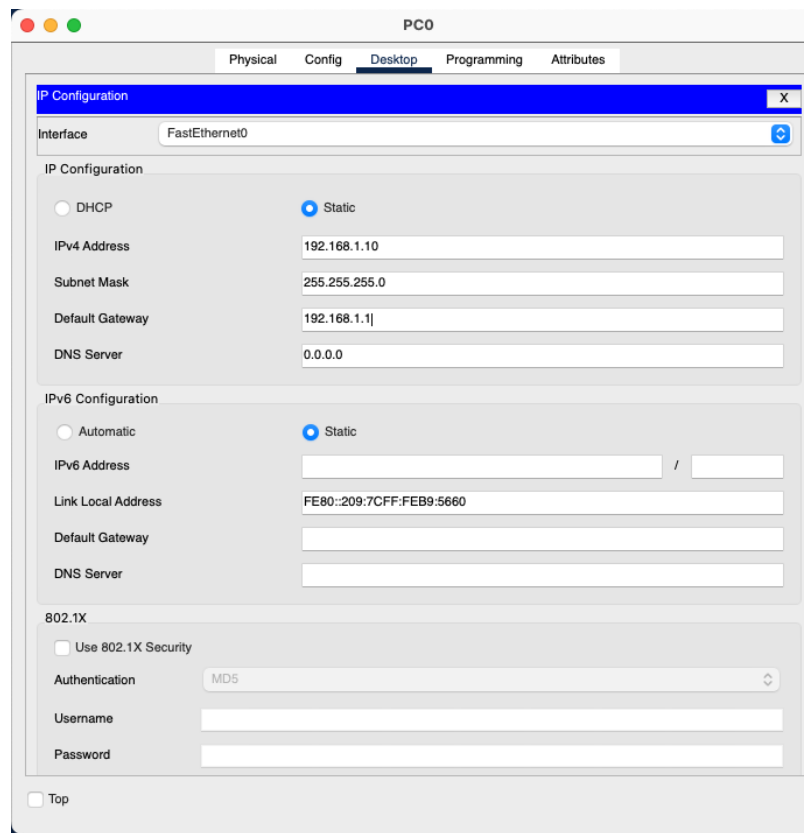


Figure 4. PC0 initial static IP configuration.

5. Protocol Demonstrations

5.1 ICMP — Connectivity Testing

ICMP (Internet Control Message Protocol) was used via ping and traceroute to verify Layer 3 reachability throughout the build process. A ping from PC0 to Router0 (192.168.1.1) confirmed the gateway was reachable with zero packet loss.

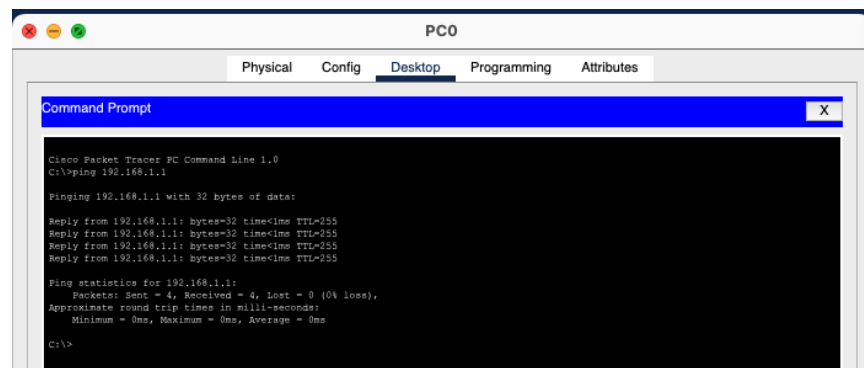


Figure 5. Successful ping from PC0 to the default gateway (4/4 packets, 0% loss).

5.2 DHCP — Automatic IP Address Assignment

Server0 was configured as a DHCP server with a pool ('serverPool') serving addresses starting at 192.168.1.10, with default gateway 192.168.1.1 and DNS server 192.168.1.2 handed out automatically to clients. PC0 and PC1 were then switched from static to DHCP mode and both received leases from the pool (192.168.1.10 and 192.168.1.11 respectively) without any manual entry.

The screenshot shows the 'Server0' configuration window with the 'Services' tab selected. On the left, a 'SERVICES' sidebar lists various services, with 'DHCP' highlighted. The main area is titled 'DHCP' and contains the following configuration fields:

- Interface: FastEthernet0 (dropdown menu)
- Service: On (radio button)
- Pool Name: serverPool
- Default Gateway: 192.168.1.1
- DNS Server: 192.168.1.2
- Start IP Address: 192, 168, 1, 10 (four separate input fields)
- Subnet Mask: 255, 255, 255, 0 (four separate input fields)
- Maximum Number of Users: 246
- TFTP Server: 0.0.0.0
- WLC Address: 0.0.0.0

Below these fields are 'Add', 'Save', and 'Remove' buttons. At the bottom, a table displays the configured DHCP pool:

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168...	192.168...	192.168...	255.255...	246	0.0.0.0	0.0.0.0

Figure 6. DHCP pool configuration on Server0.

PC0

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 192.168.1.10

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 192.168.1.2

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::209:7CFF:FEB9:5660

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

Figure 7. PC0 — 'DHCP request successful,' leased 192.168.1.10.

PC1

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 192.168.1.11

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 192.168.1.2

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2E0:F7FF:FE73:258B

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

Figure 8. PC1 — 'DHCP request successful,' leased 192.168.1.11.

5.3 DNS — Name Resolution

Server0 also runs the DNS service, with an A record mapping the hostname 'webserver.local' to Server1's address, 192.168.1.3. This allows clients to reach the web server by name instead of by IP address, which was then verified directly via a browser request (Section 4.4).

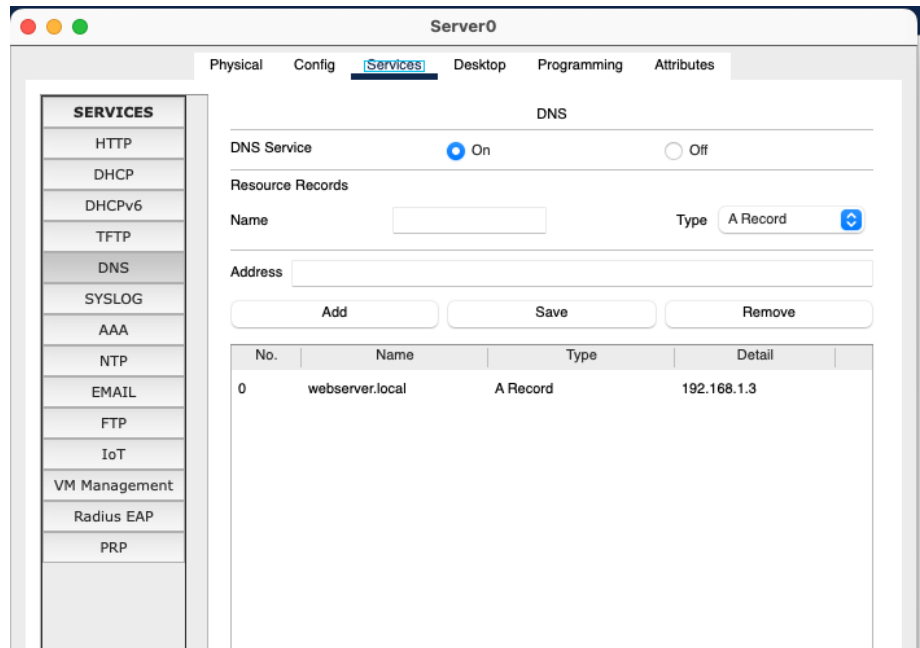


Figure 9. DNS A record: webserver.local -> 192.168.1.3.

5.4 ARP — Address Resolution

Address Resolution Protocol operates automatically beneath IP-layer communication: before any device can deliver a frame on the local network, it must resolve the destination's IP address to a MAC address. This was directly observed in Packet Tracer's Simulation Mode, where ARP request/reply packets appear as a distinct, color-coded event type preceding each new conversation between devices (see Section 5.7, Figure 15).

5.5 HTTP / HTTPS — Web Communication

Server1 was configured as a web server with both HTTP (port 80) and HTTPS (port 443) enabled. From PC1's browser, requesting 'webserver.local' triggered a DNS lookup, a TCP connection to Server1, and an HTTP GET request that returned the server's default page. The same request over HTTPS (https://webserver.local) confirmed the server also serves content securely over TLS on port 443.

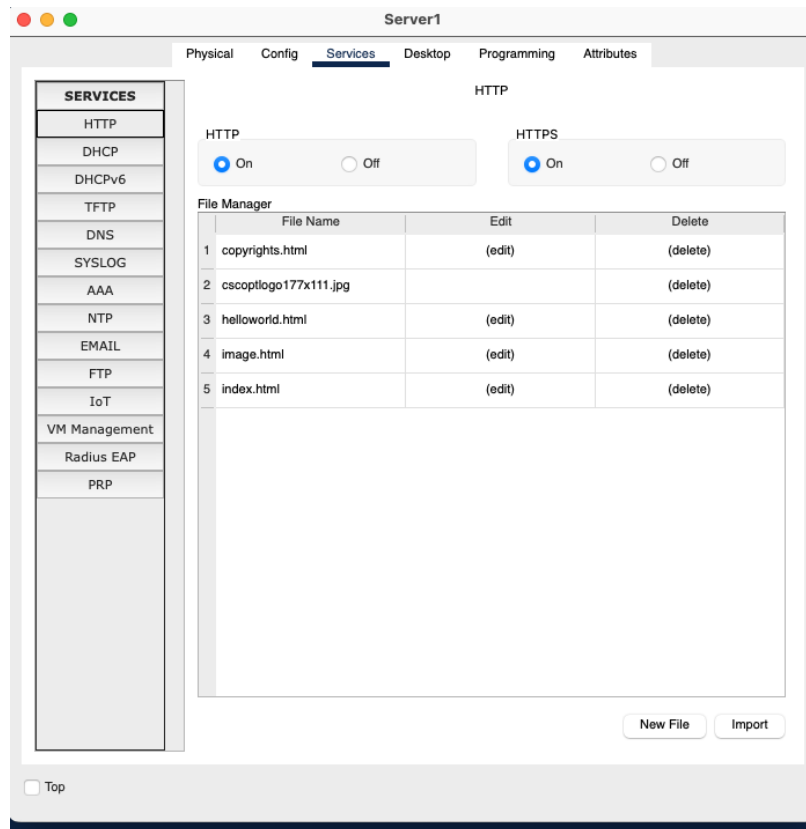


Figure 10. HTTP and HTTPS services enabled on Server1.

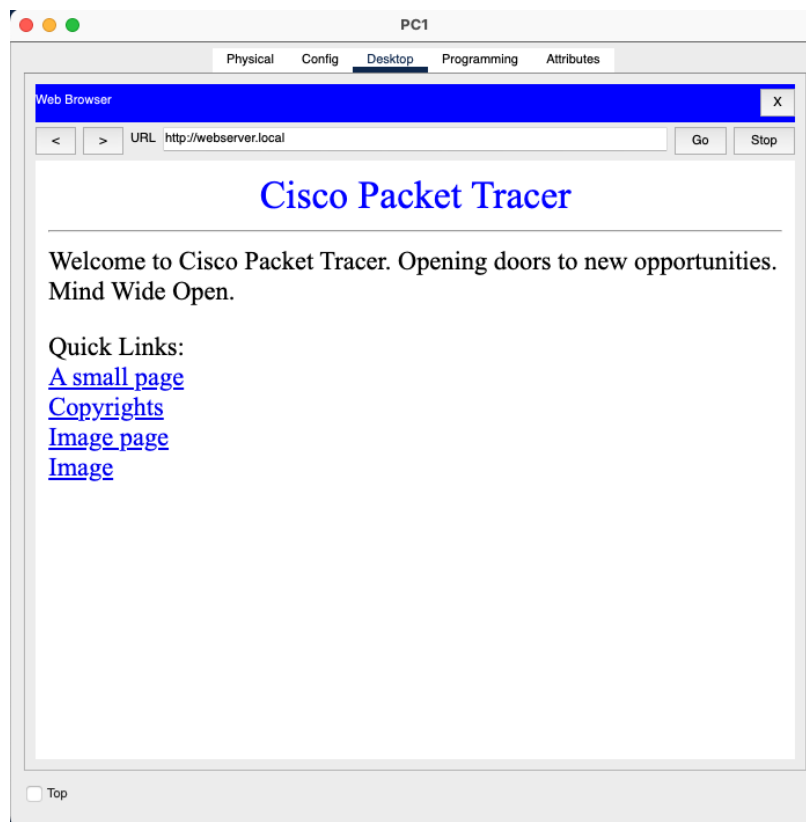


Figure 11. PC1 browser successfully loading `http://webserver.local`.

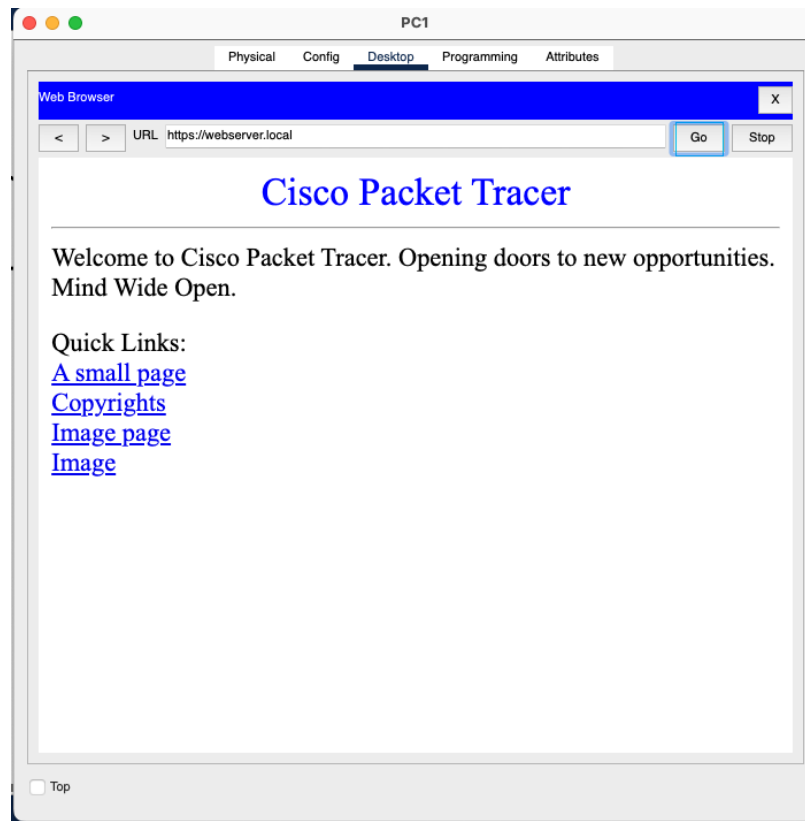


Figure 12. PC1 browser successfully loading `https://webserver.local`.

5.6 SSL/TLS — Secure Communication

The successful HTTPS page load in Figure 12 confirms Server1 accepted a TLS-secured connection on port 443. In a full packet capture, this would appear as a TLS handshake (ClientHello / ServerHello / certificate exchange) layered on top of the underlying TCP three-way handshake described in Section 5.7.

5.7 TCP / UDP — Transport Layer Behavior

Packet Tracer's Simulation Mode was used to step through the traffic generated by a browser request to `webserver.local`, with the Event List capturing every packet exchanged across the topology. This single capture surfaces multiple transport-layer behaviors at once: the initial DNS lookup (UDP, since DNS queries prioritize speed over guaranteed delivery), followed by a TCP handshake and HTTP request/response for the page itself.

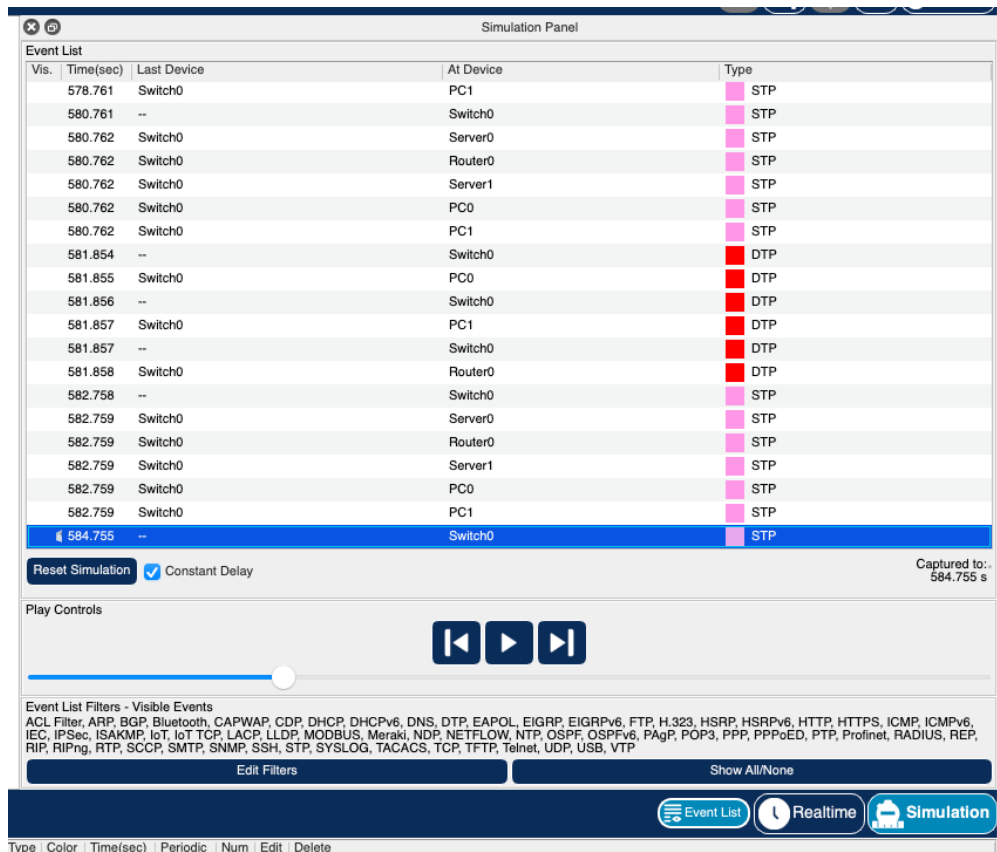


Figure 13. Simulation Mode Event List showing the sequence of packets exchanged during a client-server session.

5.8 NAT — Internal/External Communication

Router0 was configured with NAT overload (PAT) to translate all traffic sourced from the internal 192.168.1.0/24 network to its outside interface address (203.0.113.1) before forwarding it to the simulated external network. Access-list 1 defines which internal addresses are eligible for translation, and the 'overload' keyword allows multiple internal hosts to share the single outside address simultaneously — the same mechanism used by consumer routers to share one public IP address across a household.

```

Router>configure terminal
^
% Invalid input detected at '^' marker.

Router#enable
Router>configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet0/1
Router(config-if)#ip address 203.0.113.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#interface gigabitEthernet0/0
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#interface gigabitEthernet0/1
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)#ip nat inside source list 1 interface gigabitEthernet0/1 overload
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
|

```

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Figure 14. Router0 NAT configuration — inside/outside interfaces, access-list, and overload rule.

```

Pinging 203.0.113.5 with 32 bytes of data:

Request timed out.
Reply from 203.0.113.5: bytes=32 time<1ms TTL=127
Reply from 203.0.113.5: bytes=32 time<1ms TTL=127
Reply from 203.0.113.5: bytes=32 time<1ms TTL=127

Ping statistics for 203.0.113.5:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

```

Figure 15. PC0 successfully reaching the simulated external host (203.0.113.5) through NAT (3/4 replies; the first request timed out while ARP resolved the path, a normal and expected first-packet behavior).

5.9 Traceroute

A traceroute from PC0 to the web server (192.168.1.3) confirmed the path taken through the network, completing in a single hop since both devices reside on the same local subnet.

```

Cisco Packet Tracer PC Command Line 1.0
C:\>tracert 192.168.1.3

Tracing route to 192.168.1.3 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      192.168.1.3

Trace complete.

C:\>

```

Figure 16. Traceroute from PC0 to 192.168.1.3.

6. Testing and Analysis Summary

- Connectivity verified via ICMP ping between clients, servers, and the router gateway with 0% packet loss.
- DHCP successfully leased addresses to both client PCs from a defined pool, including gateway and DNS options.
- DNS successfully resolved a hostname (webserver.local) to the correct internal IP address.
- HTTP and HTTPS requests both completed successfully against the web server, confirming TCP-based application traffic and TLS-secured delivery.
- NAT overload correctly translated internal 192.168.1.0/24 traffic to the router's external address, confirmed via a successful ping to a simulated external host.
- Packet Tracer Simulation Mode captured the underlying ARP, DNS (UDP), and TCP/HTTP exchanges for a single client-server transaction.

7. Challenges Faced and Solutions Implemented

The original project proposal specified GNS3 with a Docker-based router and Linux server containers. During setup, two architecture-specific obstacles emerged:

- VirtualBox and Cisco IOSv, the originally planned router platform, are x86-based and do not run natively on Apple Silicon (M-series) Macs, which lack the virtualization support required – this is the primary reason the project moved away from GNS3 toward Cisco Packet Tracer.
- GNS3's Docker container integration requires the GNS3 VM (a Linux virtual machine) to create the network namespaces Docker needs; it cannot create Docker-based devices using a local macOS server, even with Docker/OrbStack installed and working correctly on the host.

Rather than invest further setup time into a Linux VM workaround, the project was migrated to Cisco Packet Tracer — a tool explicitly listed as an approved alternative in the assignment's technology list. Packet Tracer runs natively on Apple Silicon, includes built-in DHCP, DNS, and HTTP/HTTPS server roles, and provides a Simulation Mode capable of visualizing packet-level protocol behavior without requiring a separate packet-capture tool. The topology, IP addressing scheme, and full protocol list from the original proposal were preserved unchanged; only the underlying simulation platform changed.

8. Conclusion

This project successfully implemented a small office network demonstrating the complete set of required protocols: DHCP, DNS, ARP, HTTP/HTTPS, ICMP, TCP, UDP, NAT, SSL/TLS, and IP addressing/subnetting. Each protocol was independently tested and verified, and the overall design shows how these protocols work together in a realistic client-server environment — from a client obtaining an address via DHCP, to resolving a hostname via DNS, to establishing a secure TCP/TLS connection with a web server, all while the router transparently translates addresses for any traffic destined outside the local network.